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Deep learning based classification of facial dermatological disorders

Evgin Goceri

Department of Biomedical Engineering, Engineering Faculty, Akdeniz University, Turkey

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ABSTRACT

Common properties of dermatological diseases are mostly lesions with abnormal pattern and skin color (usually redness). Therefore, dermatology is one of the most appropriate areas in medicine for automated diagnosis from images using pattern recognition techniques to provide accurate, objective, early diagnosis and interventions. Also, automated techniques provide diagnosis without depending on location and time. In addition, the number of patients in dermatology departments and costs of dermatologist visits can be reduced. Therefore, in this work, an automated method is proposed to classify dermatological diseases from color digital photographs. Efficiency of the proposed approach is provided by 2 stages. In the 1st stage, lesions are detected and extracted by using a variational level set technique after noise reduction and intensity normalization steps. In the 2nd stage, lesions are classified using a pre-trained DenseNet201 architecture with an efficient loss function. In this study, five common facial dermatological diseases are handled since they also cause anxiety, depression and even suicide death. The main contributions provided by this work can be identified as follows: (i) A comprehensive survey about the state-of-the-art works on classifications of dermatological diseases using deep learning; (ii) A new fully automated lesion detection and segmentation based on level sets; (iii) A new adaptive, hybrid and non-symmetric loss function; (iv) Using a pre-trained DenseNet201 structure with the new loss function to classify skin lesions; (v) Comparative evaluations of ten convolutional networks for skin lesion classification. Experimental results indicate that the proposed approach can classify lesions with high performance (95.24% accuracy).

1. Introduction

Dermatological diseases are significant illnesses affecting people in all age groups (at any age) and geographies [1–3]. The authors in Ref. [4] have informed that dermatological diseases affect 1.9 billion people worldwide. A recent study has indicated that dermatological diseases are actually more widespread than we think since many people affected by these diseases do not consult a dermatologist [5]. People with dermatological disorders have higher risk for mental comorbidities (e.g., anxiety, depression, addiction) [6–10].

Particularly, people with facial dermatological diseases (such as acne vulgaris (Fig. 1a), psoriasis (Fig. 1b), rosacea (Fig. 1c), seborrheic dermatitis (Fig. 1d)) are living serious physiological and sociological problems [11–16]. Even, these patients have suicidal thoughts with a high ratio. For example, a study in Lithuania has indicated that the rate of patients having suicidal thoughts is 9.8% among 543 patients with facial dermatological diseases [12]. Similarly, a high risk of suicide attempt and suicidal death of the people with psoriasis disease in Quebec has been shown in a recent study [16].

Diagnosis of dermatological disorders relies on manual examination and visual cues. Therefore, it depends on instant visual perceptions by the naked eye and experiences of dermatologists, which may lead to misdiagnosis, delayed treatments and increase the cost of treatment. However, early diagnosis and timely interventions are required for many dermatological diseases. For example, psoriasis, which is a chronic inflammatory disease and psoriasis patients are much more likely to have metabolic syndromes compared to the general populations [17,18]. The traditional diagnosis approach is not only subjective but also can be more difficult if the disease is infectious (like *impetigo*) or the patient has another infectious disease. Because, the diagnosis should be performed without closing to the patient due to the spreading risks of the disease. In addition, there is not enough dermatologist, particularly in rural regions, and consultations are costly [1,19]. In addition, the conventional method can lead to infections and pains.

To provide accurate, early and objective diagnosis in dermatology, lesions should be recognized from images automatically. Automated classification of lesions also helps to diagnose these diseases without requiring touching the patient. In addition, they can be used without depending on location and time. Further, diagnosis with automated lesion classification is more cost-effective than the traditional diagnosis

E-mail address: evgin@akdeniz.edu.tr.

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approach. However, the automated methods proposed in the literature have some limitations or drawbacks (Section 2). Therefore, the goal in this work is to apply a novel fully automated technique to overcome these limitations and drawbacks.

The proposed approach in this work consists of 2 stages. In the 1st stage, lesions are detected and extracted with a new fully automatic level-set based technique, which is an updated version of the Automated Detection of Facial Disorders (ADFD) method [20] and called as Fully ADFD (F-ADFD). In the F-ADFD method, active contours are set automatically using a binary image which is obtained with a K-means based clustering after denoising and intensity normalization steps. In the 2nd stage, segmented lesions are classified with a deep learning-based technique, which uses the Dense convolutional Network (DenseNet201) [21] architecture as a backbone. A new loss function has been integrated into the network architecture. Because, for an image classification task, in addition to the design of a network architecture, loss functions used in classification layers have a significant role to increase feature differences between classes and also to reduce feature variations within classes as soon as possible. The flow chart of the proposed approach is presented in Fig. 2.

In this work, five common facial dermatological diseases, which are acne vulgaris, psoriasis, hemangioma, seborrheic dermatitis and rosacea, have been handled. Also, colored digital photographs, which were obtained by mobile phones or cameras, have been used. The image data sets have been provided from several databases such as DermNet [22], DermQuest [23], DermWeb [24] and Dermatoweb [25]. The main reasons to use this kind of images are the current technological advancements in phones and potential software, which can be used by dermatologists and patients at anytime/anywhere on mobile devices. The overall contributions provided by this study can be identified as follows:

(i) Presenting a comprehensive survey about the state-of-the-art works on deep network-based classifications of dermatological diseases with a focus on classifications from photographs.
(ii) Proposing a new fully automated segmentation method to detect and extract skin lesions.
(iii) Proposing a new adaptive, hybrid and non-symmetric loss function.
(iv) Using a pre-trained DenseNet201 model for lesion classification from colored photographs.
(v) Application of 9 Convolutional Neural Network (CNN) based architectures with the new loss function for lesion segmentation, and comparative performance analyses of the 10 network architectures applied in this work

Except the DenseNet201, these following 9 networks have been applied: VGGNet16 [26] and its deeper version as VGGNet19, GoogleNet [27], InceptionV3 [28], Xception [29], InceptionResNetV2 [30], ResNet18 [31] and deeper versions as ResNet50, ResNet101, and also the proposed DenseNet201.

Remaining parts are structured as follows: The F-ADFD technique and related works are explained in Section 2. Experimental results are given in Section 3. Quantitative evaluations of the results obtained from the F-ADFD method and the networks applied in this work are given in Section 4. Discussions and conclusions are presented in Section 5.

2. Related works

2.1. ADFD method [20]

The ADFD method is based on rough and fine segmentation steps. In the rough segmentation step, lesions are extracted roughly according to the intensity of RGB (Red, Green, Blue) color components. Because, facial redness is a common issue of many of facial skin diseases (Fig. 1). Redness in the image indicates that the intensities of red color components are greater when they are compared with the intensities of blue and green color components. Because of this, an image is generated with the difference of the green and red color values. Then, normalization and clustering by K-means [32] into 4 clusters is applied. The pixel values corresponding to the reddish regions are grouped in the 3rd and 4th clusters when the clustering is applied. Therefore, these two clusters are chosen to obtain an image showing roughly segmented lesions. The fine segmentation step is based on the level set technique that uses active contours defined using a Partial Differential Equation (PDE). The main advantages provided by PDEs are that solutions of PDEs are found using finite difference methods, it is possible to integrate regularization terms, also their computational costs are low.

The ADFD method uses a *variational* level set technique which performs segmentation by computing the minimum value of the following function:

$$E(\varphi) = \alpha A_g(\varphi) \quad (\alpha \in \Re) \tag{1}$$

when a contour is given by $C = \{(x,y) \in \Omega | \varphi(x,y) = 0\}$ (Here, zero level set is represented with $\varphi(x,y) = 0$and $I : \Omega \rightarrow \Re$ is an image). In (1), α is a constant value, $A_g(\varphi)$ refers to the area inside the contour and is computed using $A_g(\varphi) = \int_\Omega gH(-\varphi)dxdy$ with the Heaviside function, $H(.)$. The function $|C_p| > 0$, is used for boundary detection and is defined as $g = Spf(I_{Gray}(\mathrm{x}))\nabla Spf(I_{Gray}(\mathrm{x}))\nabla I_{s_1}(\mathrm{x})$ where the term $\nabla I_{s_1}(\mathrm{x})$ refers to the gradient of the grayscale image. The term $Spf(I_{Gray}(\mathrm{x}))$ refers to the signed pressure force function that is region-based and defined with the difference between the intensity levels of the grayscale image and the average of intensity levels inside the active contour. Evolution of the contour stops at borders since the energy function in (1) has the minimum value at boundaries. To obtain the minimum of the energy function, gradient flow is computed at each iteration by $\partial\varphi/\partial t = -\partial E/\partial\varphi$. The direction of the curve is the opposite of the *Gateaux* gradient $(\partial E/\partial\varphi)$.

The ADFD method uses both edge- and region-based knowledge at every iteration. Therefore, the method has advantages of a region-based [33–35] and an edge-based active contour model [36,37]. These models have their own disadvantages and advantages. For example, an edge-based active contour model has the ability to stop contour evolution at boundaries more efficiently, while a region-based active contour is usually less sensitive to the active contour location defined initially.



Fig. 1. Acne vulgaris (a); Psoriasis (b); Rosacea (c); Seborrheic dermatitis (d).

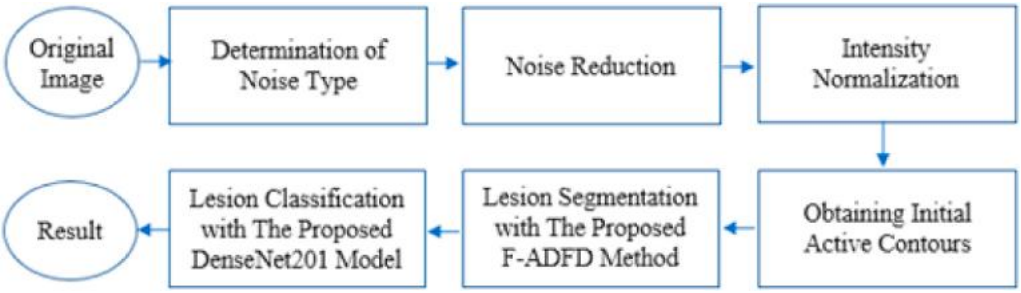


Fig. 2. Flow chart of the proposed approach.

2.2. Classification methods for dermatological diseases

In the literature, automated classification methods proposed for dermatological diseases are usually performed using dermoscopy images [38,39]. Most of them use supervised techniques. Therefore, the ground-truth data and hand-crafted features are used in the training step. The main problem of these conventional methods is their inability to generalize because of high variations, which are caused by lighting conditions, different zooming effects, dark borders, several operators, shadows, instrumentations, skin lines or light reflections in image data sets. Also, several artifacts and a limited number of images in training stages increase the generalization problem. Therefore, feature extractions that require efficient algorithms and engineering skills to detect the required features might be inefficient and lead to mis-classifications [38,40].

To obtain high accuracy in lesion classification, CNN-based techniques have been implemented to use in dermatology. The state-of-the-art studies except those approaches presented in a recent review [41] are given in Table 1 with their results. These methods have the potential to classify skin lesions. However, unlike our work, the authors mostly focused on melanoma or other types of skin cancer, plus clinical or dermoscopy images were used (Table 1). In our work, five facial dermatological diseases (acne vulgaris, psoriasis, hemangioma, seborrheic dermatitis and rosacea) have been handled. Also, colored digital photographs have been used since they can be obtained easily at anytime and anywhere with the advanced mobile phone technology which are widely used worldwide. Digital photographs are constructed with different camera and imaging techniques when they are compared to clinical and dermoscopy images. Different artifacts, resolutions, color features, lightning conditions, noise amount and types caused by

Table 1
The state-of-the art works on dermatological disease classification with deep learning.

Disease	Image Type	Network Architecture	Whole Image/Segmented Lesion	Results	Reference
Melanoma	Dermoscopy images	Region based CNN and iteration-controlled Newton-Raphson	Segmented lesion (by region-based CNN)	Accuracy with ISBI2016 and ISBI2017 is 94.5% and 93.4%, respectively	[42]
Melanoma	Dermoscopy images	CNN with LeNet		Accuracy with ISBI2016 and PH_2 data set is 75% and 97.91%, respectively	[43]
Skin cancer	Dermoscopy images	Mutual bootstrapping CNN	Segmented lesion (by a coarse segmentation network)	Average AUC with PH_2 data set and ISIC2017 is 97.7% and 93.8%, respectively	[44]
Skin cancer	Dermoscopy images	Multi-scale multi-class CNN	Segmented lesion (by cropping)	Accuracy is 86.2%	[45]
Skin cancer	Dermoscopy images	InceptionV3, ResNet50, Inception-ResNetV2, and DenseNet201	Segmented lesion (by full resolution CNN)	Accuracy with InceptionV3, ResNet50, Inception-ResNetV2, and DenseNet201 for 7 classes of ISIC2018 is 88.05%, 89.28%, 87.74%, and 88.70%, respectively	[46]
Seborrheic keratosis, melanoma and nevus	Dermoscopy images	AlexNet		Accuracy, specificity and sensitivity are 95.91%, 93.00% and 88.47% respectively	[47]
Acral Lentiginous Melanoma (ALM), intermediate tumors, acral benign nevi	Dermoscopy images	CNN		Accuracy for ALM is 74.7% in stage I and 79.0% in stage II	[48]
Melanoma	Dermoscopy images	Combination of GoogleNet, ResNet101, NasNet		Accuracy with PH_2 and ISIC2019 is 93% and 95%, respectively	[49]
Melanoma	Dermoscopy images	CNN		Mean specificity is 86.5%, mean sensitivity is 87.5%	[50]
Several	Clinical	Fine-tuned VGG19		Accuracy is 50.27%	[51]
Melanocyticnevus, SK BCC, psoriasis	Dermoscopy images	InceptionV3		Accuracy with data set-A and data set-B is $87.25 \pm 2.24\%$ and $86.63 \pm 5.78\%$, respectively	[52]
Malignant and benign skin lesions	Dermoscopy images	CNN with filters of 7×7 and 16 fully connected layers	Segmented lesion (by generalizedGaussian distribution)	Accuracy, sensitivity and specificity are 98.32%, 98.15%, 98.41%, respectively	[53]
Melanoma and Nevi	Clinical and dermoscopy images	ResNet50		Sensitivity and specificity are 89.4% and 64.4%, respectively	[54]
Melanoma, Nevi, SK, Akiec, BCC, DF, BKL	Dermoscopy images	InceptionV3		F1 score (normalized) is 0.93	[55]
Melanoma vs other	Dermoscopy images	ResNet50 and ResNet101		Accuracy with Ham1000, ISBI2017, ISBI2016 is 89.8%, 95.60% and 90.20%, respectively	[56]

different camera and imaging techniques may lead to obtain unexpected results from a method which gives successful results with some image types. Therefore, those methods proposed with clinical or dermoscopy images are not always efficient for photographs due to the different characteristics of images.

To examine those methods proposed with photographs, comprehensive research has been performed in the literature, and it has been observed that there are only a few works [57–64] and they have drawbacks. For example, in Ref. [57], morphological operations have been applied using structural elements. However, classification of different types of lesions using the same structural element with the same size produces unexpected results. In Ref. [58], a multi-layer perceptron has been applied with hand-crafted features. However, it can cause unsuccessful results if irrelevant features are used since classification depends on the chosen features. Logical regression has been proposed in Ref. [59]. However, images having different characteristics have been used since they are obtained with spectral imaging by integrating an extra camera on the mobile phone. Recently, the authors in Ref. [60,61] have applied MobileNet and the authors in Ref. [62] have used a DenseNet and ResNet structures with photographs taken by mobile phones. Although, their results indicate success of these techniques, they have addressed different kinds of diseases (skin cancer in Ref. [60,61], leukoderma, crust, erythema, ulcer, pustule, etc. in Ref. [62]) rather than the diseases addressed in this work.

In two conference proceedings, published more recently, the same facial diseases have been handled and some preliminary results obtained by different strategies have been presented [63,64]. In Ref. [63], five deep networks (U-Net, VGGNet19, ResNet50, InceptionV3, Inception-ResNetV2) have been applied using whole photographs and high performance of ResNet50 has been presented according to training accuracy and loss values. To improve the performance, in Ref. [64], three diseases have been handled, and the images showing only lesion regions rather than whole photographs have been used as inputs for these networks. Also, a deeper ResNet architecture with 101 layers has been applied. According to their results, ResNet101 produces better results (77.72% accuracy) than the others. Our work has been designed and developed to achieve higher accuracy.

2.3. Loss functions

The traditional cross-entropy is almost the default loss function that is used with softmax in the literature (Section 2) because it has a clear theory and provides easy training. However, the feature discrimination ability of this loss function is not good [65]. Although it has a potential to control False Negative (FN) (which refers to the object pixel that is wrongly labeled as background) and False Positive (FP) (which refers to the background pixel that is wrongly labeled as object), it is not efficient in case of a smaller object relative to the large background. In other words, it has the potential to control output imbalance, but it is not efficient in case of input imbalance during training [66]. The input imbalance problem particularly occurs in skin lesion classifications since the number of lesion pixels is usually less when it is compared to the number of non-lesion pixels. To deal with this problem, several loss functions have been applied. However, they have their own disadvantages. For example, contrastive [67] and triplet loss [68] functions are applied. However, they need to pay much attention to local features which cause long convergence time and training difficulties. Center loss function is proposed in Ref. [69]. It minimizes the distance between samples and class centers calculated in each batch. Then, it combines the cross-entropy and the mean-square error. However, class centers are optimized continuously during the training process, which makes it difficult to update the actual centers during training and the distances among similar classes may not be separated well. The loss function in Ref. [70] is a kind of k-nearest-neighbor algorithm, which needs large storage space and increases computational cost. Therefore, in this work, a new loss function has been proposed and tested to solve the classification problem of facial dermatological diseases.

3. Proposed approach

In the proposed approach, lesion detection, segmentation and classification are performed sequentially. In this section, details of these stages are explained.

3.1. Lesion detection and segmentation

The ADFD approach is efficient in segmentation of lesions from photographs according to the accuracy and processing time given in Ref. [20]. However, it has these limitations: (i) User defined initial contours inside of the lesions are required. However, users may define them outside or only a part of the contour might be inside of the lesions. In these cases, the ADFD method gives unacceptable results. (ii) The energy functional to be minimized includes the constant coefficient α which may not be a good value for every photograph including noise or artifacts. Also, there is not a noise reduction step in the ADFD method. Therefore, in this stage, the F-ADFD method has been developed by updating the ADFD method to overcome its limitations.

To provide robust segmentation in the proposed F-ADFD method, noise type determination and noise reduction (Section 3.1.1), contrast enhancement (Section 3.1.2), automatic initialization of active contours (Section 3.1.3) and level set evolution with adaptive coefficients (Section 3.1.4) have been performed.

3.1.1. Noise type determination and noise reduction

In the literature, to reduce different types of noise (e.g., speckle, impulse, uniform), several techniques were proposed, such as, iterative filtering [71], non-local means filtering [70], speckle filtering [72], wavelet transform based [73] or graphical model based [74] methods. To decide the most appropriate denoising method to apply to an image, the noise type in images should be determined. For this purpose, in this work, these two statistical features of noise signals have been examined: Kurtosis and skewness [75,76]. To compute these feature values, four regions (about 50 × 50 pixels) where gray level values are homogenous (or close to homogenous) are chosen from inside and outside of the lesions. The kurtosis [3–0.26, 3 + 0.61] and skewness [0.05, 0 + 0.83] computed from these regions indicate that the noise type in our images is Gaussian. Therefore, pixel based linear filtering has been applied to reduce the noise. In this method, Weiner filtering has been used with grayscale images ($I(x,y)$) by:

$$B(x,y) = \mu + \left[\left(\sigma^2 - k^2\right) / \sigma^2\right](I(x,y) - \mu) \quad (2)$$

In (2), μ (mean) and σ (variance) are computed with eight-pixel values neighbor to each pixel. The term k^2 refers to the average of local variance values. Weiner filter adapts itself to the local variance. If the variance is large (small), then less (more) flattening effect occurs. Therefore, it produces better results than the other linear filtering methods. Fig. 3 shows an original image (Fig. 3a), the image obtained after the noise reduction (Fig. 3b), the noise extracted from the original image as grayscale (Fig. 3c) and the same noise in HSV (Hue, Saturation, Value) space to increase visualization (Fig. 3d).

3.1.2. Contrast enhancement

After examination of several contrast enhancement methods proposed in the literature [77], contrast enhancement has been provided by the following algorithm to obtain efficient results:

1) Conversion of the image to La × b* space from RGB color space
2) Scaling of double-type data in images to the [0 1] range (To perform this task, the values in the luminosity channel are divided by 100 since luminosity values span a range between 0 and 100)

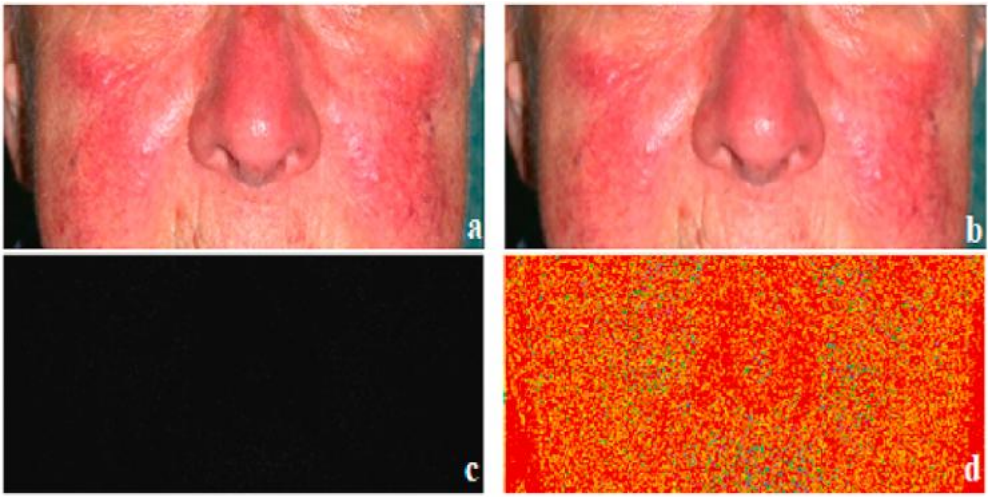


Fig. 3. Original image (a); The image after noise reduction (b); The noise extracted from the image as grayscale (c); The noise in the HSV space to increase visualization (d).

3) Saturation of pixels in the 1% region from the top and bottom side to apply contrast enhancement only to the values in the luminosity channel by keeping the values in the a × and b × channels the same. (Results are multiplied by 100 (the maximum luminosity value) again)
4) Conversion of images from La × b* space to RGB space

An example image obtained by the contrast enhancement method with the image presented in Fig. 3. b is presented in Fig. 4.

3.1.3. Automated initialization of active contours

To provide automated initialization of active contours, RGB colored images (Fig. 5a) have been used. An image has been generated (Fig. 5b) with the difference of the values in red and green color components and intensity normalization has been applied. In the normalization process, intensity values in the image (Fig. 5b) have been divided by the biggest intensity value in the same image. After the normalization, the pixels showing lesions have higher intensity values than the remaining pixels (Fig. 5c). Therefore, the image obtained by normalization has been clustered into clusters (for example, 7 clusters, which are (1) background (pixels with zero value), (2) dark gray, (3) gray, (4) bright gray, (5) bright, (6) brighter, and (7) the brightest gray level valued pixels). Then, a new binary image is generated by choosing the pixel values grouped in the last two clusters (Fig. 5d). The binary image is used to set the initial active contour function that is given as partitioned by Ref. [20];



Fig. 4. Image obtained by the contrast enhancement method.

$$\varphi_0(x) = \begin{cases} -k, & ifx \in R_0 - \partial R_0 \\ 0, & ifx \in \partial R_0 \\ k, & ifx \in R - R_0 \end{cases} \quad (3)$$

In (3), the term x refers to pixel values, k is a positive constant value ($k > 0$), R_0 refers to a subregion (the binary image in Fig. 5d) and the term ∂R_0 refers to edges of the subregion. The initial active contour expands during the evolution and stops automatically when it reaches to the boundaries of the lesion (Fig. 5e).

In this work, clustering has been performed with K-means which is a deterministic iterative technique and widely used [78]. An important disadvantage of the traditional K-means method is that clustering results vary in accuracy according to the number of clusters chosen by users. However, it is not clear how to decide the most appropriate number of clusters for an image. Even if it is found for an image, clustering of all images using the same constant number of clusters can cause unexpected results due to different gray level distributions. Therefore, in the K-means method applied in this work, the most appropriate number of clusters for each image has been obtained automatically by Elbow technique [79]. For example, the number of clusters obtained by this technique is 12 for the image shown in Fig. 5. d, while it is 8 for the image shown in Fig. 6. a. If an initial active contour has been created as multiple (Fig. 6a), then they close to each other during the evolution and merge until reaching the edges of the lesion (Fig. 6b). If separated lesions exist in an image, then the initial active contours on each lesion remain as separated during evolution, and the lesion boundaries are detected separately by these contours. (The merging, breaking or being separated properties of active contours have been explained in Ref. [80].)

3.1.4. Obtaining adaptive coefficients

The energy functional given in (3) is defined with the constant coefficient α that affects the velocity of the active contour. If a big value is assigned to it, then the contour motion becomes fast that can lead to over-segmentation. If a small value is assigned to it, then the contour motion becomes slow, and even it can stop before reaching to the lesion boundaries. In the ADFD method, its value was found experimentally. However, using the same constant coefficient for every image can cause unsuccessful results since lesions may have not only different shapes but also different pattern and color intensity. Also, the color intensity in a lesion may not be homogenous in all regions. Therefore, finding the optimal α value experimentally for each image is a challenging issue. To solve this problem, in this work, the coefficient has been computed automatically for each image by using curvature feature of the contour at every iteration during evolution.

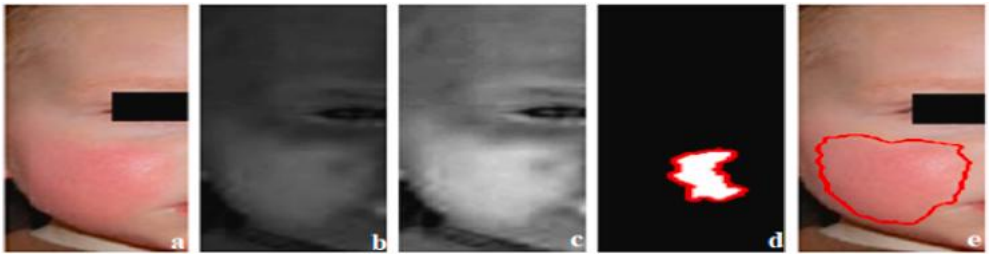


Fig. 5. Original image (a); Image generated with the difference of the values in red and green color component (b); Image obtained after intensity normalization (c); Initial active contour shown with red color (d); The active contour reached to the lesion boundaries after evolution (e).

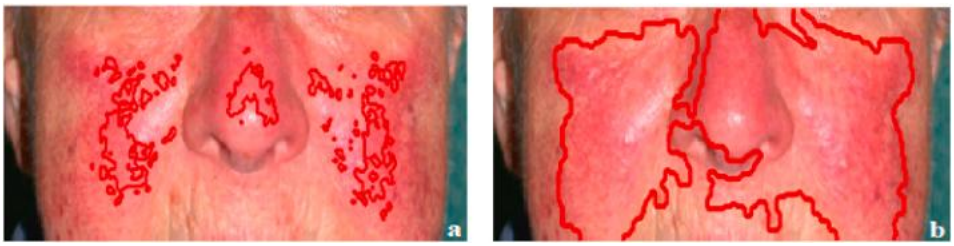


Fig. 6. Initial active contour created as multiple (a); Active contour reaching the lesion boundaries and stopping (b).

Curvature is an intrinsic feature [81]. It affects velocity of active contours. Therefore, active contour evolution with speed depending on curvatures was implemented in many works [82–86].

In Fig. 7, a contour evolution, $C_t = \kappa\vec{N}$, with time t, curvature (κ) and normal vector $(\vec{N})$ is illustrated. The contour evolution according to the minimum and maximum curvature was investigated, and the results obtained with different images have been presented in Ref. [87]. It is known that minimum curvatures allow inward concave sides to expand outward and suppress the motion of outward convex sides. Maximum curvatures allow outward sides to shrink inward and suppress the evolution in inward concave sides [87].

The mean curvature obtained by taking the average of the maximum and minimum curvature provides the motion of the contour to be smooth. In this way, a derivative of the function is taken, and the contour motion becomes fast [87–89]. For this reason, in this work, α value has been computed with the maximum and minimum curvature terms by the following equation:

$$\alpha = (|\max(\kappa)| + |\min(\kappa)|) / 2 \quad (4)$$

where κ corresponds to the curvature and obtained by $\kappa = div\left(\frac{\nabla\varphi}{|\nabla\varphi|}\right)$. Signs of these curvature terms affect the direction of the motion. In this work, the direction is outside, which means that α should be positive. Therefore, in (4), absolute values of these curvatures have been used.

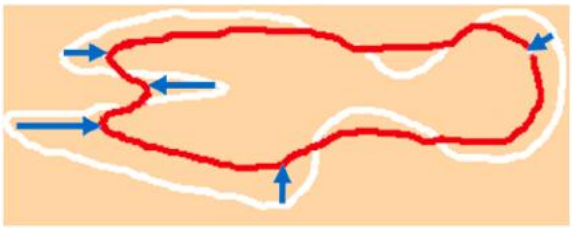


Fig. 7. Curvature effect on the motion of an active contour (initial active contour is represented with white color, the active contour after evolution is represented with red color).

3.2. Lesion classification

In this stage, classification has been performed by employing transfer learning with a CNN. Because, as explained in Ref. [90,91], after training on a large labeled dataset (e.g., ImageNet [92]), transfer learning method can be adopted. In other words, CNN has an ability to learn generic features that are applicable to other images without training from scratch. Also, initializing the weights of a CNN pre-trained on large data sets increases its accuracy for a specific task where training data is not enough [93]. In addition, transfer learning enables to speed up the learning process by reducing training time and dataset size.

The CNN architecture used as a backbone in this work is DenseNet [21] which has been trained with more than one million images from one-thousand categories in ImageNet [92] for distinct generic image descriptors. The main reasons to choose DenseNet201 in this study are (i) its proficiency in providing maximum information flow between layers, (ii) encouraging feature reuse, and (iii) strengthen feature propagation. In this architecture, each layer obtains additional features that are passed on from all preceding layers and the same layers transfer its features to all subsequent layers. As a result, deep layers will have rigorous feature maps that have lost their minimal spatial information. Another advantage of the dense connection is preventing over-fitting as it has a regularizing effect [21].

DenseNet201 architecture has been re-trained with our datasets by fine-tuning and replacing the last layer, which is the classification layer with softmax, with the one allowing to classify into 5 classes. Also, a new hybrid loss function (Section 3.3) has been integrated into this layer.

To apply the proposed DenseNet201 model, input images have been resized to (224, 224) to be appropriate to the model. Also, learning rate and epoch numbers have been set to 0.0003 and 9, respectively. Optimization has been performed by using stochastic gradient descent algorithm and activation has been provided by Rectified Linear Unit (ReLU). To avoid from any bias, the network has been applied with balanced data sets. The training set has been constructed using 80 images from each class, and the testing set has been constructed using 21 images from each class.

3.3. Loss function

An important modification in the DenseNet201 architecture is to use

a new loss function identified with the cross-entropy (C) and Tversky (Tv) similarity functions as follows:

$$L = (1-c)C + (c)Tv \tag{5}$$

The weight value c has been set as 0.3 in our work. In (5), the term C is used to control FNs and FPs. It is computed by:

$$C = p\log(q) + (1-p)\log(1-q) \tag{6}$$

where p refers to the probability of the ground truth label being zero. The term q refers to the probability of a logistic sigmoid function which is predicted as zero. The term $(1-p)\log(1-q)$ can penalize FPs since its value becomes zero when the prediction is correct. In (6), the term Tv is computed by summation over the class cl with $\sum_{cl}(1 - T_{index})$. Here, the term T_{index} refers to the Tversky index [94] given by:

$$T_{index} = \frac{\left|\widehat{P} \cap P\right|}{\left|\widehat{P} \cap P\right| + \alpha\left|\widehat{P} - P\right| + \beta\left|P - \widehat{P}\right|} \tag{7}$$

where $\widehat{P}$ and P corresponds to the predicted and ground truth masks, respectively. The term $\left|\widehat{P} \cap P\right|$ gives the quantity of correctly classified lesion pixels (TPs). The term $\left|\widehat{P} - P\right|$ gives the quantity of background pixels classified wrongly as lesion pixels (i.e. FPs). The term $\left|P - \widehat{P}\right|$ gives the quantity of lesion pixels classified wrongly as background pixels. Therefore, $\left|\widehat{P} - P\right| + \left|P - \widehat{P}\right|$ refers to the quantity of all incorrectly classified pixels. Therefore, α and β parameters can adjust the magnitude of penalty values for FNs and FPs, respectively. In this work, α and β values have been chosen as 0.7 and 0.3, respectively. It should be noted here that if these parameter values are chosen as 0.5, then Eq. (7) becomes the Dice similarity function which is also F1 score. To improve the network performance further, the proposed loss function has been defined as partitioned by:

$$F_{loss} = \begin{cases} w\log(1+|L|/\varepsilon) & \text{If } Th_1 > |L|, \\ L & \text{If } Th_2 > |L| > Th_1, \\ |L| + \left|Th_1 - w\log\left(1+\frac{Th_1}{\varepsilon}\right)\right| & \text{If } Th_2 \le |L|, \end{cases} \tag{8}$$

In (8), the terms w, Th_1, Th_2 are weight and threshold values chosen as 10, 0.1 and 0.9, respectively. The term $\left|Th_1 - w\log\left(1+\frac{Th_1}{\varepsilon}\right)\right|$, which produces a constant value, is used to obtain a continuous, smooth and differentiable loss function at Th_1. Since the logarithm function is monotonic, it provides smooth convergence. The term ε is used to limit the curvature of the non-linear part and it has been chosen as 0.5. It should be noted here that if ε is chosen as a very small number, then the network training can be unstable and may lead to gradient burst.

4. Results

Example results obtained from the segmentation stage are presented with original images (showing acne vulgaris (Fig. 8a), hemangioma (Fig. 8d), psoriasis (Fig. 8g), seborrheic dermatitis (Fig. 8j), rosacea (Fig. 8. m), final active contours stopped automatically at the boundaries (Fig. 8b,e,h,k,n) and segmented lesions (Fig. 8c,f,i,l,o) in Fig. 8.

To see the effect of the proposed loss function defined in Section 3.3, the DenseNet201 architecture has been applied separately with the traditional loss and the proposed loss function. The accuracy values obtained from the training and testing stages are given in Table 2.

5. Quantitative performance evaluations

In this section, quantitative evaluations of the results obtained from the proposed segmentation and classification approaches have been presented separately.

5.1. Evaluations for the proposed segmentation approach

The results obtained from the proposed segmentation approach have been evaluated according to accuracy, precision and specificity values. The values of these parameters have been calculated by:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{9}$$

$$Precision = \frac{TP}{TP + FP} \tag{10}$$

$$Specificity = \frac{TN}{TN + FP} \tag{11}$$

where TP (True Positive) is the number of pixels segmented correctly as foreground, FP (False Positive) is the number of pixels segmented falsely as foreground, TN (True Negative) is the number of pixels segmented correctly as background, and FN (False Negative) is the number of pixels segmented falsely as background.

The classification performance has also been evaluated using F1 score (generally called as F measure) and Matthew's Correlation Coefficient (MCC). F1 score has been computed with the following equation:

$$F1_{score} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{12}$$

MCC has been computed by:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \tag{13}$$

The reference images required at this stage have been obtained by manual segmentation of lesions. Average accuracy, precision, specificity, F1 score and MCC values are obtained as 95.97%, 98.88%, 97.00%, 90.90% and 89.93% respectively. The results for each disease handled in this work are given in Table 3.

The segmentation results have also been evaluated with Area Error Rate (AER) [95], i.e., Tanimoto coefficient, using binary images. AER has been computed by;

$$AER = \frac{I_{union} - I_{intersection}}{I_{manual}} \times 100\% \tag{14}$$

where the term I_{union} ($I_{intersection}$) refers to the image generated by union (intersection) of the reference image and the image obtained from the F-ADFD method. The term I_{manual} refers to the reference image segmented manually. AER is a very sensitive error metric which is affected significantly, even with a single pixel difference between the images obtained automatically and manually. The AER values computed for each disease are presented in Table 4. Average AER has been obtained as 9.86%.

The average processing time required for the segmentation algorithm is 9 s, which has been obtained by a computer including Intel Core i7 with 8 GB of DDR4 type RAM and 3.6 GHz CPU.

5.2. Evaluations for the proposed classification approach

Efficiency of the DenseNet201 structure has been evaluated by comparing its performance with those performances obtained by other CNN based networks. For this purpose, nine CNN-based architectures pre-trained on ImageNet have been used. They have been applied with the proposed loss function to make the comparison meaningful. Also, the same hyper-parameters and image sets have been used in the training and testing stages. All networks have been applied using Matlab (R2019b) on the same computer. Table 5 presents the list of these networks with their performances in terms of accuracy, loss and processing time values.



Fig. 8. Original images (a,d,g,j,m); Final active contours (b,e,h,k,n); Segmented lesions (c,f,i,l,o).

Table 2
Accuracy values computed by the DenseNet201 model.

Loss Function	Accuracy (%)	
	Training Stage	Testing Stage
Cross-entropy	88.01	72.29
The proposed loss function	96.32	95.24

Table 3
Accuracy, precision, specificity, F1 score and MCC values obtained from the results of the F-ADFD method for each disease.

Disease	Accuracy (%)	Precision (%)	Specificity (%)	F1 score (%)	MCC (%)
Rosacea	94.75	98.47	97.53	91.66	87.76
Acne Vulgaris	97.10	99.57	97.15	91.28	89.76
Hemangioma	95.67	99.20	96.82	85.56	91.40
Psoriasis	96.46	98.86	96.45	93.25	91.11
Seborrheic Dermatitis	95.89	98.32	97.09	92.75	89.65

Table 4
AER values for each disease.

	Acne Vulgaris	Hemangioma	Rosacea	Psoriasis	Seborrheic Dermatitis
AER (%)	9295	9067	10,289	10,121	10,571

Table 5
Accuracy, loss and processing time values obtained by the CNN-based architectures applied in this work.

Network Architecture	Accuracy Values (%)		Loss Values		Processing Time
	Training	Testing	Training	Testing	
VGG16	92.07	66.67	0.6	3.7	98 min, 31 s.
GoogleNet	93.35	70.48	0.6	3.5	18 min, 45 s
VGG19	93.76	72.38	0.6	3.5	189 min, 47 s.
Xception	80.00	72.38	1.4	3.5	136 min, 3 s
InceptionV3	88.75	75.24	1.2	3.4	75 min, 39 s.
ResNet18	95.11	76.19	0.5	3.4	20 min, 42 s.
ResNet50	90.04	77.14	0.7	3.3	56 min, 46 s.
ResNet101	95.00	80.00	0.5	3.1	101 min, 38 s.
InceptionResNetV2	93.65	83.81	0.6	2.9	129 min, 58 s.
Densenet201	96.32	95.24	0.5	0.5	102 min, 54 s.

Also, comparisons of these networks using average accuracy, precision, specificity, F1 score and MCC values have been performed. Table 6 shows these results.

Also, the performance of the proposed DenseNet201 network has been evaluated for each disease separately with the same parameters. Table 7 shows these parameter values.

6. Discussion and conclusion

An efficient method is proposed in this work to classify five common facial dermatological diseases using photographs. To obtain high performance; (i) A new fully automated level set based segmentation has been applied. (ii) Classification has been provided with the images showing only lesions rather than whole images. (iii) A pre-trained DenseNet201 model with a new loss function has been used.

It has been observed that active contours with the function used for edge detection can be applied efficiently in lesion segmentation from photographs. The edge detection function in this study is defined using the gradient of the grayscale image and a signed pressure force function. The force function is region-based and defined with the difference between intensities of the grayscale image and the average of intensities inside the active contour (Section 2.1). Active contours with the edge detector can reach the desired boundaries.

In the segmentation stage, the proposed F-ADFD method eliminates all limitations of the semi-automated ADFD method and achieves lesion segmentation with lower AER (9.86%) when it is compared to the AER (10.31%) of the ADFD method. When the performance of the F-ADFD method is analyzed according to those five diseases separately, it is seen that hemangioma lesions are segmented with higher performance than the other lesions (Table 4). This is mainly due to their relatively more homogenous intensities and simpler shapes. According to the numeric values given in Table 3, the F-ADFD segmentation technique is efficient according to accuracy, precision and specificity values.

Although deep networks require large data sets, an efficient technique to train these networks using small datasets is to pretrain the model on a larger dataset and then to transfer the weights to the current model. By this way, it is possible to set better initial values for the early convolutional layers. This significantly improves the network's generalization ability. In this work, transfer learning is applied to speed up the learning and to obtain high performance. Difficulties in the stages of model training and design have been eliminated using transfer learning.

The proposed adaptive, hybrid and non-symmetric loss function uses advantages of both cross-entropy and Tversky similarity functions. According to the numerical results (Table 2), the accuracy obtained using the proposed loss function are greater than the values obtained using the cross-entropy function in the same DenseNet201 architecture.

Comparative evaluations indicate that the DenseNet201 architecture produces results with the minimum loss (0.5) and maximum accuracy (95.24%) (Table 5). Also, the architecture is superior than the other 9 architectures according to precision, specificity, MCC and F1 score values (Table 6). The second highest performance has been obtained by the InceptionResNetV2 architecture. Also, it has been observed that among the ResNet architectures with different depth, the ResNet model

Table 6
Average accuracy precision, specificity, F1 score and MCC values obtained by the results of the CNN-based architectures applied in this work.

Network Architecture	Accuracy (%)	Precision (%)	Specificity (%)	F1 score (%)	MCC (%)
VGG16	66.67	67.44	91.67	65.21	58.41
GoogleNet	70.48	70.93	92.62	69.80	63.19
VGG19	72.38	93.10	93.10	69.81	64.82
Xception	72.38	71.62	93.10	71.48	65.05
InceptionV3	75.24	74.81	93.81	74.70	68.80
ResNet18	76.19	77.83	94.05	76.10	70.87
ResNet50	77.14	79.46	94.29	77.07	72.43
ResNet101	80.00	82.94	95.00	77.33	75.37
InceptionResNetV2	83.81	84.93	95.95	83.05	80.04
DenseNet201	95.24	95.37	98.80	95.05	93.78

Table 7
Performance evaluation of the proposed DenseNet201 model for each disease.

Disease	Accuracy (%)	Precision (%)	Specificity (%)	F1 score (%)	MCC (%)
Acne Vulgaris	95.23	100	100	97.56	97.01
Hemangioma	99.05	100	100	99.15	99.05
Psoriasis	98.50	95.45	98.81	97.67	97.11
Rosacea	87.81	94.47	98.81	90.00	87.12
Seborrheic Dermatitis	95.63	86.95	96.42	90.90	88.65

with 101 layers can classify lesions with higher performance than the others. When these networks are compared with respect to processing time, results show that the DenseNet201 architecture requires longer processing time than the VGG16, GoogleNet, InceptionV3 and the three ResNet architectures. However, it requires less processing time than the VGG19, InceptionResNetV2 and Xception architectures.

It has been observed that deep neural network based techniques not only have the ability to extract features automatically, but also they can integrate and enrich with different level features (low/middle/high level) by increasing the depth. They can keep relationships between data and learn patterns. Therefore, they have the ability to classify skin lesions.

Declaration of competing interest

None declared.

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